Assignment nmaps

TASK 1

```
Command Prompt

Microsoft Windows [Version 10.0.26200.8037]
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C:\Users\We.Link01>nmap -v
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-06 16:52 +0300
Read data files from: C:\Program Files (x86)\Nmap
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.47 seconds
           Raw packets sent: 0 (0B) | Rcvd: 0 (0B)

C:\Users\We.Link01>
```

Nmap is an open source tool used for

- Network discovery
- Identifying active hosts detecting open ports
- Identifying running services
- Performing security auditing

Purpose of nmaps

- Helps administrators monitor networks
- Detects vulnerabilities
- Improve system security

TASK 2

```
C:\Users\We.Link01>nmap -sn 192.168.1.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-06 17:07 +0300
Nmap scan report for 192.168.1.0
Host is up (0.037s latency).
Nmap scan report for RTK_GW (192.168.1.1)
Host is up (0.048s latency).
Nmap scan report for 192.168.1.2
Host is up (0.043s latency).
Nmap scan report for 192.168.1.3
Host is up (0.043s latency).
Nmap scan report for 192.168.1.4
Host is up (0.043s latency).
Nmap scan report for 192.168.1.5
Host is up (0.042s latency).
Nmap scan report for 192.168.1.6
Host is up (0.041s latency).
Nmap scan report for 192.168.1.7
Host is up (0.041s latency).
Nmap scan report for 192.168.1.8
Host is up (0.040s latency).
Nmap scan report for 192.168.1.9
Host is up (0.040s latency).
Nmap scan report for 192.168.1.10
Host is up (0.039s latency).
Nmap scan report for 192.168.1.11
Host is up (0.048s latency).
Nmap scan report for 192.168.1.12
Host is up (0.065s latency).
```

Active hosts are marked host is up

TASK3

```
Host is up (0.077s latency).
Nmap scan report for 192.168.1.255
Host is up (0.073s latency).
Nmap done: 256 IP addresses (256 hosts up) scanned in 12.41 seconds

C:\Users\We.Link01>nmap 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-06 17:16 +0300
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00045s latency).
Not shown: 996 closed tcp ports (reset)
PORT    STATE SERVICE
135/tcp open  msrpc
445/tcp open  microsoft-ds
902/tcp open  iss-realsecure
912/tcp open  apex-mesh

Nmap done: 1 IP address (1 host up) scanned in 0.68 seconds

C:\Users\We.Link01>nmap -p 1-1000 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-06 17:17 +0300
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00036s latency).
Not shown: 995 closed tcp ports (reset)
PORT    STATE    SERVICE
135/tcp open     msrpc
137/tcp filtered netbios-ns
445/tcp open     microsoft-ds
902/tcp open     iss-realsecure
912/tcp open     apex-mesh

Nmap done: 1 IP address (1 host up) scanned in 1.90 seconds
```

First command scans default ports

Second command scans ports 1-1000

And we are able to identify open ports

PORT STATE SERVICE

135/tcp open msrpc

137/tcp filtered netbios-ns

445/tcp open Microsoft-ds

902/tcp open iss-realsecure

912/tcp open apex-mesh

TASK 4

Service detection

```
C:\Users\We.Link01>nmap -sV 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-06 17:25 +0300
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00020s latency).
Not shown: 996 closed tcp ports (reset)
PORT    STATE SERVICE         VERSION
135/tcp open  msrpc           Microsoft Windows RPC
445/tcp open  microsoft-ds?
902/tcp open  ssl/vmware-auth VMware Authentication Daemon 1.10 (Uses VNC, SOAP)
912/tcp open  vmware-auth     VMware Authentication Daemon 1.0 (Uses VNC, SOAP)
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 8.90 seconds
```

TASK 5

OS detection

```
C:\Users\We.Link01>nmap -0 127.0.0.1
nmap: illegal option -- 0
See the output of nmap -h for a summary of options.

C:\Users\We.Link01>nmap --0 127.0.0.1
nmap: unrecognized option `--0'
See the output of nmap -h for a summary of options.

C:\Users\We.Link01>nmap -h
Nmap 7.99 ( https://nmap.org )
Usage: nmap [Scan Type(s)] [Options] {target specification}
TARGET SPECIFICATION:
  Can pass hostnames, IP addresses, networks, etc.
  Ex: scanme.nmap.org, microsoft.com/24, 192.168.0.1; 10.0.0-255.1-254
  -iL <inputfilename>: Input from list of hosts/networks
  -iR <num hosts>: Choose random targets
  --exclude <host1[,host2][,host3],...>: Exclude hosts/networks
  --excludefile <exclude_file>: Exclude list from file
HOST DISCOVERY:
  -sL: List Scan - simply list targets to scan
  -sn: Ping Scan - disable port scan
  -Pn: Treat all hosts as online -- skip host discovery
  -PS/PA/PU/PY[portlist]: TCP SYN, TCP ACK, UDP or SCTP discovery to given ports
  -PE/PP/PM: ICMP echo, timestamp, and netmask request discovery probes
  -PO[protocol list]: IP Protocol Ping
  -n/-R: Never do DNS resolution/Always resolve [default: sometimes]
  --dns-servers <serv1[,serv2],...>: Specify custom DNS servers
  --system-dns: Use OS's DNS resolver
  --traceroute: Trace hop path to each host
SCAN TECHNIQUES:
  -sS/sT/sA/sW/sM: TCP SYN/Connect()/ACK/Window/Maimon scans
  -sU: UDP Scan
```

```
  --traceroute: Trace hop path to each host
SCAN TECHNIQUES:
  -sS/sT/sA/sW/sM: TCP SYN/Connect()/ACK/Window/Maimon scans
  -sU: UDP Scan
  -sN/sF/sX: TCP Null, FIN, and Xmas scans
  --scanflags <flags>: Customize TCP scan flags
  -sI <zombie host[:probeport]>: Idle scan
  -sY/sZ: SCTP INIT/COOKIE-ECHO scans
  -sO: IP protocol scan
  -b <FTP relay host>: FTP bounce scan
PORT SPECIFICATION AND SCAN ORDER:
  -p <port ranges>: Only scan specified ports
    Ex: -p22; -p1-65535; -p U:53,111,137,T:21-25,80,139,8080,S:9
  --exclude-ports <port ranges>: Exclude the specified ports from scannin
  -F: Fast mode - Scan fewer ports than the default scan
  -r: Scan ports sequentially - don't randomize
  --top-ports <number>: Scan <number> most common ports
  --port-ratio <ratio>: Scan ports more common than <ratio>
SERVICE/VERSION DETECTION:
  -sV: Probe open ports to determine service/version info
  --version-intensity <level>: Set from 0 (light) to 9 (try all probes)
  --version-light: Limit to most likely probes (intensity 2)
  --version-all: Try every single probe (intensity 9)
  --version-trace: Show detailed version scan activity (for debugging)
SCRIPT SCAN:
  -sC: equivalent to --script=default
  --script=<Lua scripts>: <Lua scripts> is a comma separated list of
           directories, script-files or script-categories
  --script-args=<n1=v1,[n2=v2,...]>: provide arguments to scripts
  --script-args-file=filename: provide NSE script args in a file
  --script-trace: Show all data sent and received
  --script-updatedb: Update the script database.
  --script-help=<Lua scripts>: Show help about scripts.
```

```
  --badsum: Send packets with a bogus TCP/UDP/SCTP checksum
OUTPUT:
  -oN/-oX/-oS/-oG <file>: Output scan in normal, XML, s|<rIpt kIddi3,
     and Grepable format, respectively, to the given filename.
  -oA <basename>: Output in the three major formats at once
  -v: Increase verbosity level (use -vv or more for greater effect)
  -d: Increase debugging level (use -dd or more for greater effect)
  --reason: Display the reason a port is in a particular state
  --open: Only show open (or possibly open) ports
  --packet-trace: Show all packets sent and received
  --iflist: Print host interfaces and routes (for debugging)
  --append-output: Append to rather than clobber specified output files
  --resume <filename>: Resume an aborted scan
  --noninteractive: Disable runtime interactions via keyboard
  --stylesheet <path/URL>: XSL stylesheet to transform XML output to HTML
  --webxml: Reference stylesheet from Nmap.Org for more portable XML
  --no-stylesheet: Prevent associating of XSL stylesheet w/XML output
MISC:
  -6: Enable IPv6 scanning
  -A: Enable OS detection, version detection, script scanning, and traceroute
  --datadir <dirname>: Specify custom Nmap data file location
  --send-eth/--send-ip: Send using raw ethernet frames or IP packets
  --privileged: Assume that the user is fully privileged
  --unprivileged: Assume the user lacks raw socket privileges
  -V: Print version number
  -h: Print this help summary page.
EXAMPLES:
  nmap -v -A scanme.nmap.org
  nmap -v -sn 192.168.0.0/16 10.0.0.0/8
  nmap -v -iR 10000 -Pn -p 80
SEE THE MAN PAGE (https://nmap.org/book/man.html) FOR MORE OPTIONS AND EXAMPLES

C:\Users\We.Link01>
```

TASK 6

Advanced scans

```
C:\Users\We.Link01>nmap -sS 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-06 17:37 +0300
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0012s latency).
Not shown: 996 closed tcp ports (reset)
PORT    STATE SERVICE
135/tcp open  msrpc
445/tcp open  microsoft-ds
902/tcp open  iss-realsecure
912/tcp open  apex-mesh

Nmap done: 1 IP address (1 host up) scanned in 0.71 seconds

C:\Users\We.Link01>nmap -sU 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-06 17:38 +0300
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0012s latency).
Not shown: 994 closed udp ports (port-unreach)
PORT     STATE         SERVICE
137/udp  open|filtered netbios-ns
1900/udp open|filtered upnp
4500/udp open|filtered nat-t-ike
5050/udp open|filtered mmcc
5353/udp open|filtered zeroconf
5355/udp open|filtered llmnr

Nmap done: 1 IP address (1 host up) scanned in 46.88 seconds

C:\Users\We.Link01>
```

```
C:\Users\We.Link01>nmap -A 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-06 17:41 +0300
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0011s latency).
Not shown: 996 closed tcp ports (reset)
PORT    STATE SERVICE         VERSION
135/tcp open  msrpc           Microsoft Windows RPC
445/tcp open  microsoft-ds?
902/tcp open  ssl/vmware-auth VMware Authentication Daemon 1.10 (Uses VNC, SOAP)
912/tcp open  vmware-auth     VMware Authentication Daemon 1.0 (Uses VNC, SOAP)
Device type: general purpose
Running: Microsoft Windows 11
OS CPE: cpe:/o:microsoft:windows_11
OS details: Microsoft Windows 11 24H2 - 25H2
Network Distance: 0 hops
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
| smb2-time:
|   date: 2026-04-06T14:42:01
|_  start_date: N/A
| smb2-security-mode:
|   3.1.1:
|_    Message signing enabled and required

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 25.47 seconds

C:\Users\We.Link01>
```

Syn fast and stealthy

UDP slow but finds UDP services

Aggressive most detailed but noisy

TASK 7

Vulnerability scan

```
C:\Users\We.Link01>nmap --script vuln 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-06 17:47 +0300
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00045s latency).
Not shown: 996 closed tcp ports (reset)
PORT    STATE SERVICE
135/tcp open  msrpc
445/tcp open  microsoft-ds
902/tcp open  iss-realsecure
912/tcp open  apex-mesh

Host script results:
|_smb-vuln-ms10-054: false
|_smb-vuln-ms10-061: Could not negotiate a connection:SMB: Failed to receive bytes: ERROR
|_samba-vuln-cve-2012-1182: Could not negotiate a connection:SMB: Failed to receive bytes: ERROR

Nmap done: 1 IP address (1 host up) scanned in 27.58 seconds
```

The system has important Windows services exposed (especially port 445).

- SMB-related vulnerability checks failed due to connection issues, not necessarily because vulnerabilities exist.
- The result FALSE for MS10-054 means:

The system is not vulnerable to that specific exploit.

- Errors suggest:
- SMB service may be restricted
- Firewall may be blocking detailed probing

TASK 8

Other commands

```
C:\Users\We.Link01>nmap -F 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-06 18:06 +0300
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00049s latency).
Not shown: 98 closed tcp ports (reset)
PORT    STATE SERVICE
135/tcp open  msrpc
445/tcp open  microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 0.81 seconds

C:\Users\We.Link01>nmap -Pn 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-06 18:07 +0300
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00053s latency).
Not shown: 996 closed tcp ports (reset)
PORT    STATE SERVICE
135/tcp open  msrpc
445/tcp open  microsoft-ds
902/tcp open  iss-realsecure
912/tcp open  apex-mesh

Nmap done: 1 IP address (1 host up) scanned in 0.68 seconds

C:\Users\We.Link01>
```

```
C:\Users\We.Link01>nmap -T4 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-06 18:09 +0300
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00093s latency).
Not shown: 996 closed tcp ports (reset)
PORT    STATE SERVICE
135/tcp open  msrpc
445/tcp open  microsoft-ds
902/tcp open  iss-realsecure
912/tcp open  apex-mesh

Nmap done: 1 IP address (1 host up) scanned in 0.65 seconds
```